INTRODUCCIÓ A ADA

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ÍNDEX

IF /THEN /ELSE

```
if condicio then
    Codi;
else
    codi ;
end if;
```

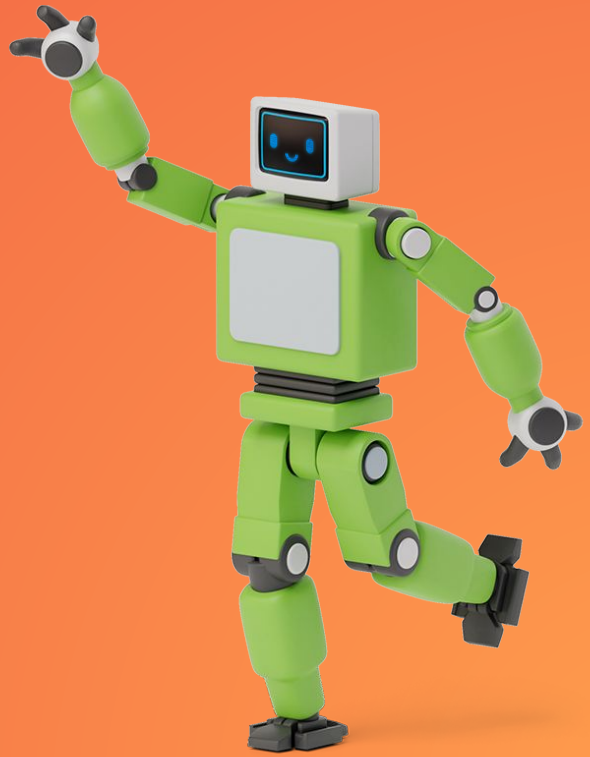
IF /THEN /ELSE

```ada
procedure Check_Positive is
   N : Integer;
begin
   --  Put a String
   Put ("Enter an integer value: ");

   --  Reads in an integer value
   Get (N);

   --  Put an Integer
   Put (N);

   if N > 0 then
      Put_Line (" is a positive number");
   else
      Put_Line (" is not a positive number");
   end if;
end Check_Positive;
```



IF /THEN /ELSE

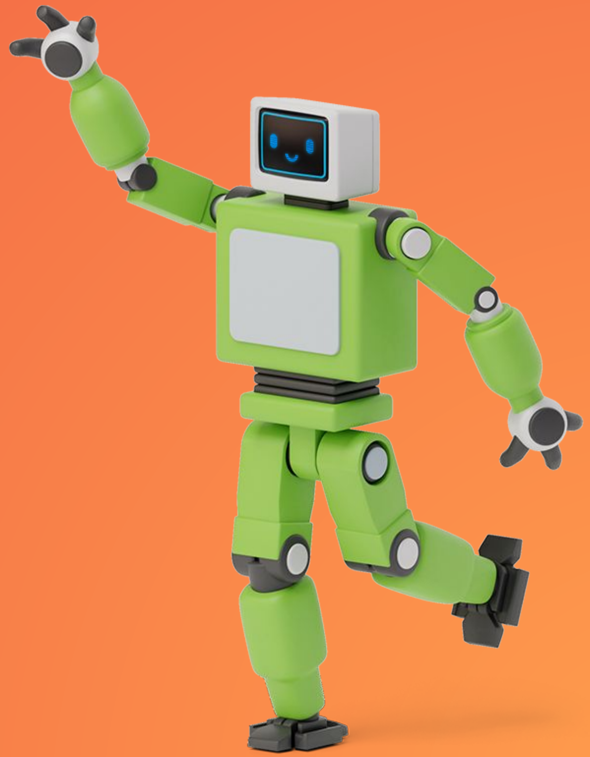
COM ENGANXAM DIVERSES CONDICIONS?

```
if condicio then
    Codi;
elseif condicio then
    codi ;
else
    codi ;
end if;
```

IF /THEN /ELSE

```
procedure Check_Direction is
   N : Integer;
begin
   Put ("Enter an integer value: ");
   Get (N);
   Put (N);

   if N = 0 or N = 360 then
      Put_Line (" is due north");
   elsif N in 1 .. 89 then
      Put_Line (" is in the northeast quadrant");
   elsif N = 90 then
      Put_Line (" is due east");
   elsif N in 91 .. 179 then
      Put_Line (" is in the southeast quadrant");
   elsif N = 180 then
      Put_Line (" is due south");
   elsif N in 181 .. 269 then
      Put_Line (" is in the southwest quadrant");
   elsif N = 270 then
      Put_Line (" is due west");
   elsif N in 271 .. 359 then
      Put_Line (" is in the northwest quadrant");
   else
      Put_Line (" is not in the range 0..360");
   end if;
end Check_Direction;
```



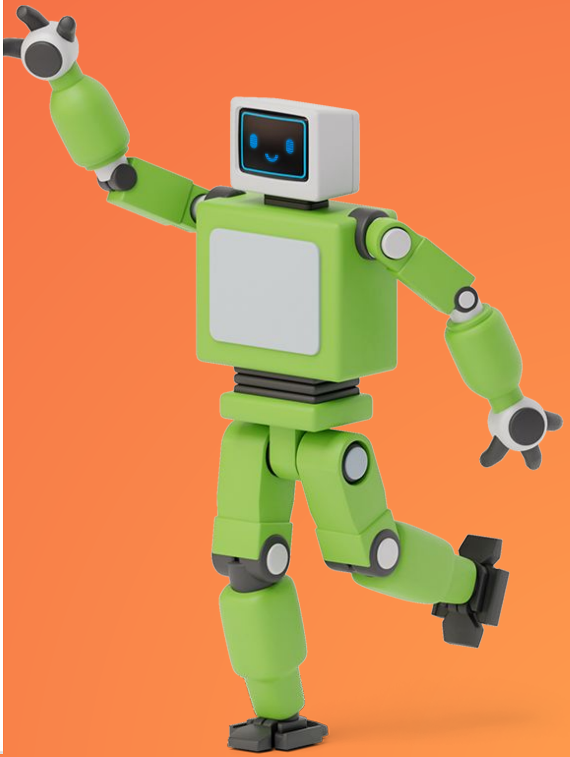
CASE

```
case condicio is
    when x =>
        codi ;
    when y =>
        codi ;
    when others =>
        Codi;
        exit;
    end case;
```

CASE

```
procedure Check_Direction is
   N : Integer;
begin
   loop
      Put ("Enter an integer value: ");
      Get (N);
      Put (N);

      case N is
         when 0 | 360 =>
            Put_Line (" is due north");
         when 1 .. 89 =>
            Put_Line (" is in the northeast quadrant");
         when 90 =>
            Put_Line (" is due east");
         when 91 .. 179 =>
            Put_Line (" is in the southeast quadrant");
         when 180 =>
            Put_Line (" is due south");
         when 181 .. 269 =>
            Put_Line (" is in the southwest quadrant");
         when 270 =>
            Put_Line (" is due west");
         when 271 .. 359 =>
            Put_Line (" is in the northwest quadrant");
         when others =>
            Put_Line (" Au revoir");
            exit;
      end case;
   end loop;
end Check_Direction;
```



EXPRESIONES CONDICIONALES

```
procedure Check_Positive is
   N : Integer;
begin
   Put ("Enter an integer value: ");
   Get (N);
   Put (N);

   declare
      S : constant String :=
        (if N > 0 then " is a positive number"
         else " is not a positive number");
   begin
      Put_Line (S);
   end;
end Check_Positive;
```

EXPRESIONES CONDICIONALES

```
procedure Main is
begin
   for I in 1 .. 10 loop
      Put_Line (if I mod 2 = 0 then "Even" else "Odd");
   end loop;
end Main;
```

```
procedure Main is
begin
   for I in 1 .. 10 loop
      Put_Line (case I is
                  when 1 | 3 | 5 | 7 | 9 => "Odd",
                  when 2 | 4 | 6 | 8 | 10 => "Even");
   end loop;
end Main;
```

FOR LOOP

```
for I in x...y loop
      codi ;
end loop;
```

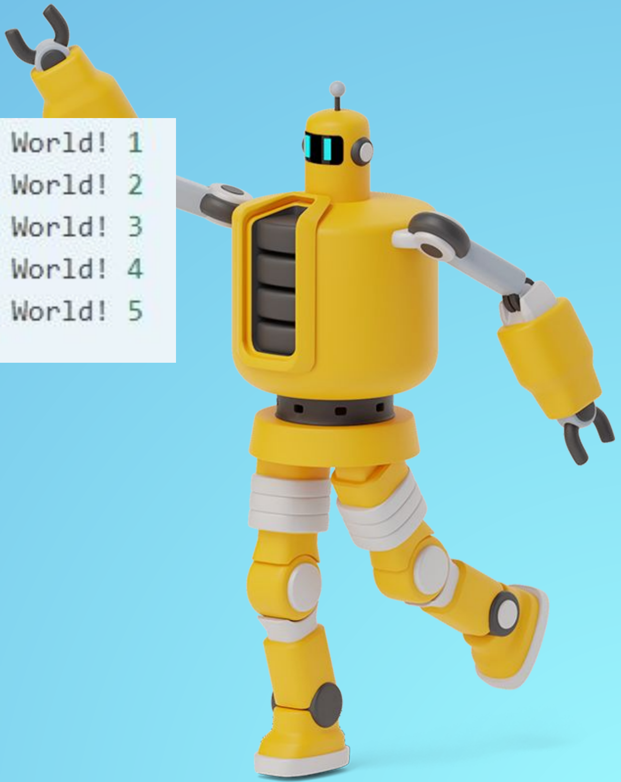
FOR LOOP

```
procedure Greet_5a is
begin
   for I in 1 .. 5 loop
      --  Put_Line is a procedure call
      Put_Line ("Hello, World!" & Integer'Image (I));
      --          ^ Procedure parameter
   end loop;
end Greet_5a;
```

```
Hello, World! 1
Hello, World! 2
Hello, World! 3
Hello, World! 4
Hello, World! 5
```

```
procedure Greet_5a_Reverse is
begin
   for I in reverse 1 .. 5 loop
      Put_Line ("Hello, World!"
                & Integer'Image (I));
   end loop;
end Greet_5a_Reverse;
```

```
Hello, World! 5
Hello, World! 4
Hello, World! 3
Hello, World! 2
Hello, World! 1
```



FOR LOOP

- Els límits del bucle(1 i 5 de l'exemple anterior) son inclosos dins el bucle.
- I és una variable local del bucle.
- I s'incrementa en cada iteració. Intentar modificar el valor de I produirà un error de compilació.
- ATENCIÓ: Si el valor del límit superior és menor que el valor del límit inferior, el bucle mai s'executarà. EL mateix passa amb els reverse bucles.

BARE LOOP

```
loop
    codi ;
exit when condicio;
end loop;
```

BARE LOOP

```
procedure Greet_5b is
   --  Variable declaration:
   I : Integer := 1;
   --  ^ Type
   --             ^ Initial value
begin
   loop
      Put_Line ("Hello, World!"
                & Integer'Image (I));

      --  Exit statement:
      exit when I = 5;
      --          ^ Boolean condition

      --  Assignment:
      I := I + 1;
      --  There is no I++ short form to
      --  increment a variable
   end loop;
end Greet_5b;
```

WHILE LOOP

```
While condicio loop
      codi ;
end loop;
```

WHILE LOOP

```
procedure Greet_5c is
   I : Integer := 1;
begin
   --  Condition must be a Boolean value
   --  (no Integers).
   --  Operator "<=" returns a Boolean
   while I <= 5 loop
      Put_Line ("Hello, World!"
                & Integer'Image (I));

      I := I + 1;
   end loop;
end Greet_5c;
```

FUNCIONS I PROCEDIMIENTS

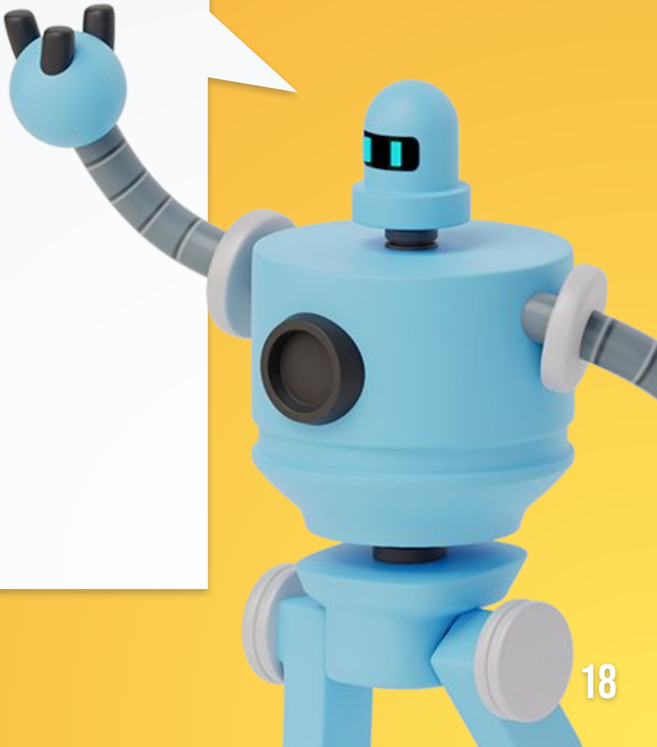
Hi ha dos tipus de subprogrames en ADA: **FUNCIONS I PROCEDIMENTS**.

DECLARACIÓ D'UNA FUNCIÓ D'EXEMPLE

```
function Increment (I : Integer) return Integer;
```

DEFINICIÓ D'UNA FUNCIÓ D'EXEMPLE

```
function Increment (I : Integer) return Integer is
   -- We define the Increment function
begin
    return I + 1;
end Increment;
```

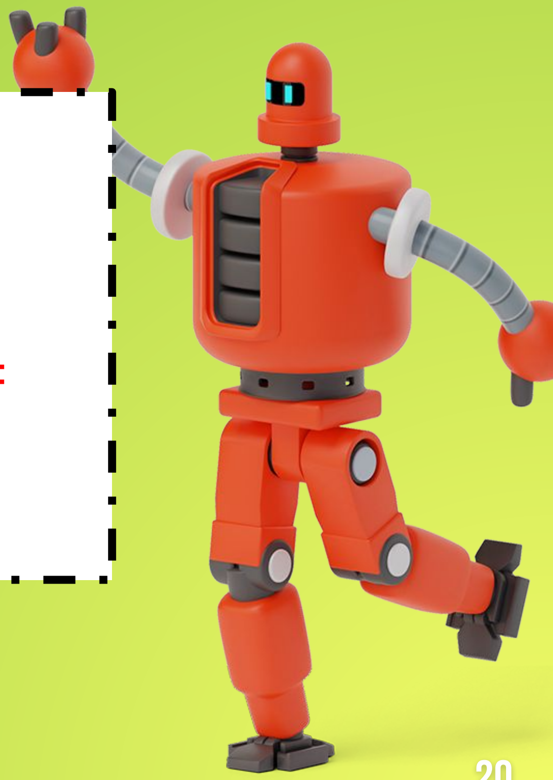


DEFINICIÓ D'UN PROCEDIMENT

```
procedure Nom_procediment is
"DECLARACIÓ VARIABLES";
begin
Codi;
end Nom_procediment;
```

ARRAYS

array (*Tipo_Índice*) of *Tipo_Elemento*
EXEMPLE:
type *Contador_Caracteres* is array (*Character*) of *Natural*;



ARRAYS

subtype *Subtipo_Índice* is *Tipo_Índice* range *Primero .. Último;*
array (*Subtipo_Índice*) of *Tipo_Elemento*;
FORMES BREUS:

- array (*Tipo_Índice* range *Primero .. Último*) of *Tipo_Elemento;*
- array *(Primero .. Último)* of *Tipo_Elemento*

ARRAYS

EXEMPLE:

```
type Contador_Caracteres is array (Character range 'A' .. 'Z') of Natural;
```

STRINGS

- Tipus especial d'Array
- Cap String en Ada pot començar pel caràcter 0.
- No tenen perquè acabar en Nul, poden tenir caràcters Nul intercalats.
- Nombre : String (1 .. 8); -- Explícitamente
- Nombre : String := "Fulanito"; --Implícitamente
- Nombre_Completo:constant String :=Nombre & ' ' & Apellidos;

CONTADOR DE PARAULES

```ada
with Ada.Text_IO, Ada.Strings.Fixed,
        Ada.Strings.Bounded, Ada.Strings.Unbounded; use Ada.Text_IO, Ada.Strings
            .Fixed, Ada.Strings.Unbounded;
procedure ComptarParaules is
    --Variable que conta paraules
    C: Integer:=0;
    --Variable que conta index dins string
    N: Integer:=1;

begin
    --Les frases introduïdes han de contenir un sentinella
    Put_line("Introdueix el text acabat amb '\' i contaré les seves paraules: ");
    declare
    --Obtenim text per teclat
    Frase: String:=Get_Line;
    begin
    --Mentre no s'hagi trobat final contam paraules
    while Frase(N) /='\' loop
    if (Frase(N) =' 'and Frase(N+1)/=' ') or Frase(N+1)='\' then
        C:= C + 1;
    end if;
    N:= N + 1;
    end loop;
    --RESULTAT
    Put_line("La frase " & Frase & " té:" & Integer'Image(C) & " paraules");
    end;
end ComptarParaules;
```

RESULTAT

```
Introdueix el text acabat amb '\' i contaré les seves paraules:
EN UN lugar de la mancha     de cuyo   nombre \
La frase EN UN lugar de la mancha     de cuyo   nombre \ té: 9 paraules
```

BIBLIOGRAFIA

https://learn.adacore.com/courses/intro-to-ada/chapters/standard_library_strings.html

https://es.wikibooks.org/wiki/Programaci%C3%B3n_en_Ada/Tipos/Arrays

https://es.wikibooks.org/wiki/Programaci%C3%B3n_en_Ada/Tipos/Strings

PLANTILLA obtinguda a :
https://www.slidescarnival.com/

```
with Ada.Text_IO, Ada.Strings.Fixed,
        Ada.Strings.Bounded, Ada.Strings.Unbounded; use Ada.Text_IO, Ada.Strings.Fixed,
Ada.Strings.Unbounded;
procedure ComptarParaules is
    --Variable que conta paraules
    C: Integer:=0;
    --Variable que conta index dins string
    N: Integer:=1;

begin
   --Les frases introduïdes han de contenir un sentinella
   Put_line("Introdueix el text acabat amb '\' i contaré les seves paraules: ");
    declare
    --Obtenim text per teclat
    Frase: String:=Get_Line;
    begin
   --Mentre no s'hagi trobat final contam paraules
   while Frase(N) /='\' loop
   if (Frase(N) =' 'and Frase(N+1)/=' ') or Frase(N+1)='\' then
      C:= C + 1;
   end if;
   N:= N + 1;
   end loop;
   --RESULTAT
   Put_line("La frase " & Frase & " té:" & Integer'Image(C) & " paraules");
   end;
end ComptarParaules;
```